

Computer Network

Complete MCQ-Oriented Notes

IBPS SO (IT OFFICER)
COMPLETE PREPARATION SERIES

100% SYLLABUS COVERAGE
CONCEPT-FOCUSED LEARNING
MCQ-ORIENTED APPROACH
QUICK REVISION READY

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Computer Network — Complete Notes

This guide deconstructs the IBPS SO (IT Officer) Computer Network syllabus to the atomic level: every layer, PDU, protocol, port, formula, and examiner trap. Concepts build progressively from beginner definitions to advanced numericals.

1. Introduction to Computer Networks

1.1 Basics of Networking

1.1.1 Definition and Goals

A **computer network** is a collection of **autonomous** computing devices (nodes) interconnected by a single transmission technology to **share resources** and **exchange information**.

The four primary goals are **resource sharing** (hardware/software/data), **high reliability** (replication/alternate paths), **cost saving / scalability**, and **communication medium** (email, chat, video).

Key fact: the devices must be **autonomous** (no master-slave forced control) and **interconnected** (able to exchange data) to qualify as a network.

1.1.2 Data Communication Components

Data communication is the exchange of data between two devices via a transmission medium. It has **5 components**.

Component	Role
Message	The information/data to be communicated
Sender	Device that sends the message (source)
Receiver	Device that receives the message (destination)
Transmission medium	Physical path (cable/air) carrying the message
Protocol	Set of rules governing the communication

Memory Trick: "M-S-R-T-P" = Message, Sender, Receiver, Transmission medium, Protocol. Without a common **protocol**, two connected devices cannot understand each other.

1.1.3 Network Criteria (Performance Metrics)

The three core criteria of a good network are **performance**, **reliability**, and **security**.

- Performance** — measured by **throughput** (actual data rate) and **delay/latency**; depends on number of users, medium, hardware, and software.
- Reliability** — measured by **frequency of failure**, **recovery time**, and **robustness** in a catastrophe.
- Security** — protecting data from **unauthorized access**, ensuring data **integrity**, and recovery from breaches/data loss.

MCQ Trap: Throughput is the actual measured rate (always $\leq$ bandwidth), while **bandwidth** is the theoretical maximum capacity. They are NOT the same.

1.1.4 Data Flow (Transmission Modes)

There are **3 modes** of data flow between two devices.

Mode	Direction	Example
Simplex	One-way only	Keyboard, Monitor, TV broadcast
Half-Duplex	Both ways, one at a time	Walkie-talkie, CB radio
Full-Duplex	Both ways simultaneously	Telephone, mobile call

SIMPLEX A -----> B (one direction)

HALF-DUPLEX A <===== or =====> B (one direction at a time)

FULL-DUPLEX A <-----> B (both at once)

Memory Trick: Full-duplex doubles capacity by giving each direction its own channel/sub-band; "Duplex = Double".

1.2 Network Types by Scale

1.2.1 Classification by Geographical Span

Type	Full Form	Span	Example
PAN	Personal Area Network	~10 m	Bluetooth, USB, hotspot
LAN	Local Area Network	Room / building (up to ~1-2 km)	Office Ethernet, Wi-Fi
CAN	Campus Area Network	Multiple buildings	University network
MAN	Metropolitan Area Network	City (up to ~50 km)	Cable TV, city Wi-Fi
WAN	Wide Area Network	Country / globe	The Internet
GAN	Global Area Network	Worldwide	Global satellite networks

- **LAN** is **privately owned**, has the **highest data rate** and **lowest error rate**; uses Ethernet/Wi-Fi.
- **WAN** spans large areas, is usually owned by **multiple organizations / ISPs**, and has higher latency and error rates.
- **VPN (Virtual Private Network)** is a **secure tunnel** built over a public network (Internet) using encryption — logically private, physically public.
- An **internetwork (internet)** connects two or more networks via routers; the global one is the **Internet**.

MCQ Trap: Order by increasing size: **PAN < LAN < CAN < MAN < WAN < GAN**. The **Internet** is the largest WAN, not a LAN.

1.3 Network Topologies

1.3.1 Connection Types

- **Point-to-Point** — a **dedicated link** between exactly two devices (entire capacity reserved for them).
- **Multipoint (multidrop)** — a **single link shared** by more than two devices (capacity shared spatially or temporally).

1.3.2 The Six Topologies

BUS
A -- B -- C -- D
|
backbone

STAR
A B C
 \ | /
 (HUB)
 / | \
D E F

RING
A --- B
| |
D --- C

MESH (full)
every node to
every other

TREE
root
/ \
hub hub

HYBRID = mix of
two or more
topologies

Topology	Cabling	Fault Tolerance	Key Formula / Note
Bus	Single backbone + terminators	Low (backbone break kills all)	Needs 2 terminators

Topology	Cabling	Fault Tolerance	Key Formula / Note
Star	Each node to central hub/switch	Hub failure kills all; node fails alone	n cables, n ports
Ring	Each node to 2 neighbours	One break can break ring (dual ring helps)	Uses token passing
Mesh	Every node to every node	Highest (multiple paths)	$n(n-1)/2$ links, n-1 ports/node
Tree	Hierarchical from root	Root/branch failure isolates subtree	Star-of-stars
Hybrid	Combination	Depends on design	e.g., star-bus

MCQ Trap: For **MESH** with **n** nodes the number of duplex links is **$n(n-1)/2$** and each node needs **(n-1)** I/O ports. For 5 nodes: **$5(4)/2 = 10$** links.

Memory Trick: "Mesh = **Maximum** links & **Maximum** reliability"; "Star dies if the **center** dies."

1.4 Network Models / Modes

1.4.1 Client-Server vs Peer-to-Peer

Feature	Client-Server	Peer-to-Peer (P2P)
Roles	Dedicated server serves clients	Every node is both client and server
Control	Centralized	Decentralized
Cost / Scale	Costly server, scales well	Cheap, scales poorly
Example	Web (HTTP), Email	BitTorrent, blockchain

1.4.2 Connection-Oriented vs Connectionless

- Connection-oriented** — a **logical connection** is established before data transfer (handshake), data is **ordered and reliable**; analogy = **telephone**. Example: **TCP**.
- Connectionless** — each packet is sent **independently** with no prior setup; **no guarantee** of order/delivery; analogy = **postal letter**. Example: **UDP, IP**.

MCQ Trap: Connection-oriented service is associated with **virtual-circuit** packet switching and **TCP**; connectionless with **datagram** switching and **UDP/IP**.

2. Reference Models

2.1 Layered Architecture

2.1.1 Why Layering

Networks are built as a **stack of layers**, each offering services to the layer above and using services of the layer below, so that complexity is **divided** (separation of concerns), and any layer can be modified **independently**.

- Protocol** — rules for communication between **peer layers** on different machines (horizontal).
- Interface** — defines services a layer offers to the layer **above** on the same machine (vertical).
- Service** — set of primitives a layer provides to the layer above.

2.1.2 PDU, Encapsulation, Decapsulation

A **Protocol Data Unit (PDU)** is the unit of data exchanged at a given layer. As data goes **down** the stack each layer adds its **header** (and the DLL adds a **trailer**) — this is **encapsulation**. Going **up** the stack each header is removed — **decapsulation**.

```

DATA  --(App)--> [H] DATA ]      (Application message)
          --(Tpt)--> [H] segment ] (Transport: SEGMENT)
          --(Net)--> [H] packet ]  (Network: PACKET / DATAGRAM)

```

```
--(DLL)--> [H] frame [T]      (Data Link: FRAME, header+trailer)
--(Phy)-->  1 0 1 1 0 0 ...  (Physical: BITS)
```

Layer	PDU Name
Application / Presentation / Session	Data / Message
Transport	Segment (TCP) / Datagram (UDP)
Network	Packet (also called Datagram)
Data Link	Frame
Physical	Bit

Memory Trick: PDU top-to-bottom = "**Some People Fear Birthdays**" → Segment, Packet, Frame, Bit (Transport→Physical). Or "**Do Some People Fear Birthdays**" adding Data at top.

2.2 OSI Reference Model

2.2.1 Overview

The **OSI (Open Systems Interconnection)** model by **ISO** (1984) has **7 layers**. It is a **reference/theoretical** model. Layers 1-3 are **media/network-support** layers, Layer 4 is the **bridge**, Layers 5-7 are **user-support** layers.

7	Application	<- user data	(HTTP, FTP, DNS, SMTP)
6	Presentation	translation/encrypt	(JPEG, ASCII, SSL)
5	Session	dialog control	(NetBIOS, RPC)
4	Transport	process-to-process	(TCP, UDP) PDU=Segment
3	Network	host-to-host route	(IP, ICMP) PDU=Packet
2	Data Link	hop-to-hop frames	(Ethernet) PDU=Frame
1	Physical	bits on medium	(cables) PDU=Bit

2.2.2 Functions of Each Layer

- **Layer 1 — Physical:** transmits raw **bits** over the medium; defines **voltage levels, data rate, pin layout, topology, transmission mode**, bit synchronization. Deals with **mechanical and electrical** specs.
- **Layer 2 — Data Link:** node-to-node (**hop-to-hop**) delivery; performs **framing, physical (MAC) addressing, error control, flow control**, and **access control**. Sublayers: **LLC** and **MAC**.
- **Layer 3 — Network:** source-to-destination (**host-to-host**) delivery across networks; performs **logical (IP) addressing, routing, fragmentation**, and **congestion control**.
- **Layer 4 — Transport:** **process-to-process** delivery; performs **segmentation/reassembly, port addressing (service-point), connection control, end-to-end flow & error control**.
- **Layer 5 — Session:** establishes, manages, and terminates **sessions (dialogs)**; provides **synchronization (checkpoints)** and **dialog control**.
- **Layer 6 — Presentation:** **translation, encryption/decryption**, and **compression**; deals with **syntax and semantics** of data.
- **Layer 7 — Application:** provides the **user interface** and network services (mail, file transfer, web).

Memory Trick: Bottom-up = "**Please Do Not Throw Sausage Pizza Away**" (Physical, Data Link, Network, Transport, Session, Presentation, Application). Top-down = "**All People Seem To Need Data Processing**".

MCQ Trap: **Encryption and compression** are at the **Presentation** layer (6), NOT Application. **Routing** is at **Network** (3), NOT Data Link. **Flow & error control** appear at BOTH Data Link (hop) and Transport (end-to-end).

2.2.3 Addressing per Layer

Layer	Address Used
Application	Name / URL
Transport	Port number (16-bit)
Network	Logical / IP address (32-bit IPv4)
Data Link	Physical / MAC address (48-bit)
Physical	None (bits)

2.3 TCP/IP Reference Model

2.3.1 The Layers

The **TCP/IP** model (DoD/ARPANET) is the **practical** model the Internet runs on. The original model has **4 layers**; a common hybrid teaching model uses **5**.

TCP/IP (4-layer)	Maps to OSI	Key Protocols
Application	Application + Presentation + Session	HTTP, FTP, SMTP, DNS, SNMP
Transport	Transport	TCP, UDP
Internet	Network	IP, ICMP, IGMP, ARP
Network Access / Link	Data Link + Physical	Ethernet, Wi-Fi, ARP

- **Application layer** — combines OSI top three; handles all high-level protocols.
- **Transport layer** — end-to-end delivery via **TCP** (reliable) or **UDP** (unreliable).
- **Internet layer** — defines the **IP** datagram and handles **routing**; best-effort, connectionless.
- **Network Access layer** — physical transmission and framing.

MCQ Trap: **ARP** is often placed at the boundary of Internet/Network-Access. In the OSI sense ARP straddles **Layer 2/3** (resolves IP→MAC). The TCP/IP model has **no separate Session and Presentation layers**.

2.4 OSI vs TCP/IP Comparison

Feature	OSI Model	TCP/IP Model
Layers	7	4 (or 5 hybrid)
Developed by	ISO	DARPA / DoD
Type	Reference / theoretical	Practical / implemented
Approach	Protocol-independent standard first	Protocol-specific (built around TCP/IP)
Transport delivery	Both CO and CL	TCP=CO, UDP=CL
Session/Presentation	Separate layers	Merged into Application
Network layer delivery	Connection-oriented & connectionless	Connectionless only (IP)

MCQ Trap: OSI's network layer supports **both** connection-oriented and connectionless service, whereas TCP/IP's Internet (IP) layer is **purely connectionless**. The "**TCP/IP**" model is named after its two main protocols.

3. Physical Layer and Data Transmission

3.1 Data and Signals

3.1.1 Analog vs Digital

- **Analog signal** — **continuous** values over time (e.g., voice, sine wave); infinite levels.
- **Digital signal** — **discrete** values (e.g., 0 and 1; square wave); finite levels.

- **Periodic signal** — repeats over a fixed **period (T)**; **frequency $f = 1/T$** .
- **Aperiodic / non-periodic** — does not repeat in a measurable pattern.

A composite signal is a combination of sine waves; **Fourier analysis** decomposes it. **Bandwidth = highest frequency – lowest frequency** (for composite signal range).

3.1.2 Bit Rate vs Baud Rate

- **Bit rate** — number of **bits** transmitted per second (bps).
- **Baud rate (signal rate)** — number of **signal units / symbols** per second.
- Relation: **Bit rate = Baud rate $\times \log_2(L)$** = baud $\times$ bits-per-symbol, where **L** = number of signal levels.

MCQ Trap: **Bit rate $\geq$ Baud rate** always. They are equal only when each symbol carries **1 bit** ($L=2$). Bandwidth is measured in **Hz**, data/bit rate in **bps**.

3.1.3 Channel Capacity — Nyquist and Shannon

Formula	Channel	Equation
Nyquist Bit Rate	Noiseless	$C = 2 \times B \times \log_2(L)$ bps
Shannon Capacity	Noisy	$C = B \times \log_2(1 + S/N)$ bps

- **B** = bandwidth (Hz), **L** = number of signal levels, **S/N** = signal-to-noise ratio (linear), **C** = capacity (bps).
- **SNR(dB) = $10 \times \log_{10}(S/N)$** .

Example: $B = 3000$ Hz, $S/N = 3162$ (≈ 35 dB)
 Shannon: $C = 3000 \times \log_2(1 + 3162)$
 $= 3000 \times \log_2(3163)$
 $\approx 3000 \times 11.63 \approx 34,881$ bps

MCQ Trap: Use **Nyquist** when the question gives **levels (L)** and says noiseless; use **Shannon** when it gives **S/N** (noisy). Shannon's formula has **no L term**. The practical capacity = **min(Nyquist, Shannon)**.

3.2 Transmission Impairments

Signals degrade as they travel. The **3 impairments** are **attenuation**, **distortion**, and **noise**.

- **Attenuation** — **loss of energy/amplitude**; compensated by **amplifiers**. Measured in **decibels (dB)**.
- **Distortion** — change in **shape** of the signal; different frequency components arrive with different delays.
- **Noise** — unwanted signals: **thermal**, **induced**, **crosstalk**, and **impulse** noise.

Memory Trick: "**A-D-N**" → Amplitude loss, shape Distortion, unwanted Noise.

3.3 Data Transmission

3.3.1 Parallel vs Serial

- **Parallel** — multiple bits sent **simultaneously** over **n wires**; fast but costly, limited distance (skew). Used inside computers (old buses).
- **Serial** — bits sent **one after another** over a **single wire**; cheaper, longer distance. Used for inter-device links (USB).

3.3.2 Serial Sub-types

Type	Synchronization	Note
Asynchronous	Start & stop bits per byte; gaps allowed	Byte-oriented, cheap, slower
Synchronous	Clock-synced bit streams , no start/stop	Frame-oriented, fast
Isochronous	Fixed timing , constant data rate, no jitter	Real-time audio/video

MCQ Trap: **Asynchronous** transmission adds **start (0) and stop (1)** bits around each character and has variable gaps; **Synchronous** has **no** start/stop bits. **Isochronous** guarantees a **fixed/constant** rate (used in streaming).

3.4 Encoding and Modulation

There are **4 conversion categories**. Memorize what each converts.

Conversion	Examples	Layer Term
Digital → Digital	NRZ-L, NRZ-I, RZ, Manchester , Diff. Manchester	Line coding
Digital → Analog	ASK, FSK, PSK, QAM	Modulation
Analog → Digital	PCM , Delta Modulation	Digitization
Analog → Analog	AM, FM, PM	Modulation

3.4.1 Digital-to-Digital (Line Coding)

- **NRZ-L (Non-Return-to-Zero-Level)** — voltage **level** represents bit; problem of **synchronization** in long runs.
- **NRZ-I (Invert)** — **transition** (inversion) at start of a **1-bit**; no change for 0.
- **RZ (Return-to-Zero)** — signal returns to **zero** mid-bit; needs **3 levels**; good sync but uses more bandwidth.
- **Manchester** — **transition in the middle** of each bit (sync + data combined). IEEE: **low→high = 1**, high→low = 0. Used in **Ethernet (802.3)**.
- **Differential Manchester** — always a **mid-bit transition for clock**; **transition at the start = 0**, no start transition = 1. Used in **Token Ring (802.5)**.

MCQ Trap: **Manchester** = Ethernet, transition encodes data + clock. **Differential Manchester** = Token Ring, presence/absence of a **start-of-bit** transition encodes data. Both are **self-clocking** (self-synchronizing).

3.4.2 Analog-to-Digital — PCM

Pulse Code Modulation (PCM) has **3 steps**: **(1) Sampling** (PAM), **(2) Quantization**, **(3) Encoding** into bits.

- **Nyquist Sampling Theorem:** sampling rate **$fs \geq 2 \times f_{max}$** (at least twice the highest frequency) to reconstruct the signal.
- **Delta Modulation (DM)** — sends only **1 bit per sample** indicating up/down change; simpler than PCM.

MCQ Trap: Telephone voice (max 4 kHz) is sampled at **8000 samples/sec** ($= 2 \times 4 \text{ kHz}$), 8 bits each → **64 kbps** PCM channel.

3.4.3 Digital-to-Analog

Scheme	Varies	Note
ASK	Amplitude	Susceptible to noise
FSK	Frequency	Better noise immunity
PSK	Phase	Used in modems; e.g., BPSK, QPSK
QAM	Amplitude + Phase	Highest data rate (e.g., 16-QAM, 64-QAM)

Memory Trick: "A-F-P-Q" vary Amplitude, Frequency, Phase, and (Amplitude+Phase). **QAM** packs the most bits per symbol.

3.4.4 Analog-to-Analog

- **AM (Amplitude Modulation)** — carrier amplitude varies (AM radio).
- **FM (Frequency Modulation)** — carrier frequency varies (FM radio).
- **PM (Phase Modulation)** — carrier phase varies.

3.5 Multiplexing and Spreading

Multiplexing lets **multiple signals share one medium**. The device is a **MUX**; reverse is **DEMUX**.

Technique	Domain	Used For
FDM	Frequency bands (analog)	Radio, TV, cable
TDM	Time slots (digital)	T1/E1 lines, telephony
WDM	Wavelengths of light	Fiber optics

- **FDM** — each signal gets a different **frequency band**, separated by **guard bands**.
- **TDM** — each signal gets a **time slot** in rotation.
- **Synchronous TDM** — fixed slots even if a source is idle (wastes bandwidth).
- **Statistical (asynchronous) TDM** — slots allocated **on demand**, more efficient.
- **WDM** — conceptually FDM for **optical** signals; combines beams of different wavelengths via a **prism**.

3.5.1 Spread Spectrum

- **FHSS (Frequency Hopping Spread Spectrum)** — carrier **hops** among many frequencies in a pseudo-random sequence.
- **DSSS (Direct Sequence Spread Spectrum)** — each bit is replaced by a **chip code** (e.g., Barker sequence), spreading the bandwidth.

MCQ Trap: **WDM** is for **fiber/optical** (wavelength = frequency for light), **FDM** for general analog. **Statistical TDM** allocates slots dynamically and is more bandwidth-efficient than **synchronous TDM**.

4. Transmission Media

Transmission media are the **physical paths** carrying signals; they belong to the **Physical layer**. Two broad classes: **guided (wired)** and **unguided (wireless)**.

4.1 Guided (Wired) Media

4.1.1 Twisted Pair Cable

Two insulated copper wires **twisted** together to reduce **electromagnetic interference (crosstalk)**.

- **UTP (Unshielded Twisted Pair)** — no shield, cheap, common in LANs; uses **RJ-45** connector.
- **STP (Shielded Twisted Pair)** — metal foil shield, less noise, costlier.

Category	Max Speed	Use
Cat 3	10 Mbps	Old telephone/Ethernet
Cat 5	100 Mbps	Fast Ethernet
Cat 5e	1 Gbps	Gigabit Ethernet
Cat 6	1-10 Gbps	Gigabit / short 10G
Cat 6a / Cat 7	10 Gbps	10G Ethernet

4.1.2 Coaxial Cable

A central copper conductor surrounded by insulation, a braided metal shield, and an outer jacket. Higher bandwidth than twisted pair; uses **BNC** connector. Types: **RG-59** (cable TV), **RG-58** (thin Ethernet), **RG-6**.

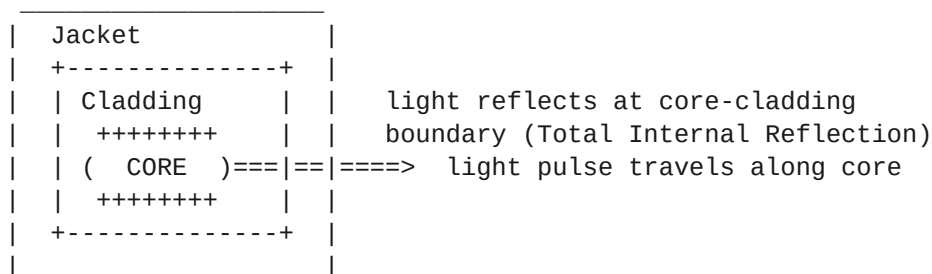
4.1.3 Fiber Optic Cable

Carries data as **pulses of light** through a glass/plastic **core** using **total internal reflection**. Immune to EMI, very high bandwidth, low attenuation, hard to tap (secure).

Type	Core	Source	Distance
Single-Mode (SMF)	~8-10 μm (narrow)	Laser	Long (km, backbones)
Multi-Mode (MMF)	~50-62.5 μm (wide)	LED	Shorter (campus/LAN)

Connectors: **SC, ST, LC, FC**. Propagation modes: **multimode step-index, multimode graded-index, single-mode**.

Fiber cross-section:



MCQ Trap: Single-mode uses a **laser** and a **narrow** core for **long** distances; multi-mode uses an **LED** and a **wider** core for shorter distances. Fiber is immune to **EMI** and offers the **highest bandwidth & lowest attenuation**.

4.2 Unguided (Wireless) Media

Signals are broadcast through **air/vacuum** (no physical conductor).

Band	Frequency	Propagation	Use
Radio waves	3 kHz - 1 GHz	Omnidirectional	AM/FM radio, cordless
Microwaves	1 - 300 GHz	Unidirectional (line-of-sight)	Cellular, satellite, Wi-Fi
Infrared	300 GHz - 400 THz	Line-of-sight, cannot pass walls	TV remote, IrDA

- Microwaves** need **line-of-sight**; **terrestrial** (towers) and **satellite** (GEO/MEO/LEO) types.
- Infrared** cannot penetrate walls → good for short-range, secure indoor links.

MCQ Trap: Radio waves are **omnidirectional**; microwaves are **unidirectional** and need line-of-sight. **Infrared cannot penetrate walls**, so it can't be used for long-range or through-wall communication.

4.3 Media Comparison

Property	Twisted Pair	Coaxial	Fiber Optic
Cost	Lowest	Medium	Highest
Bandwidth/Speed	Low-Medium	Medium	Highest
Attenuation	High	Medium	Lowest
EMI immunity	Low	Medium	Highest (immune)
Security	Low	Medium	Highest

5. Switching

Switching moves data between many devices without a full mesh, using **switching nodes**. Three techniques: **circuit**, **message**, **packet** switching.

5.1 Switching Techniques

5.1.1 Circuit Switching

A **dedicated physical path** is reserved end-to-end for the whole session via **3 phases: (1) connection setup, (2) data transfer, (3) teardown**. Example: traditional **telephone (PSTN)**.

- Fixed bandwidth, **no store-and-forward** delay during transfer, but **wastes capacity** when idle.
- Implemented by **space-division** or **time-division** switches.

5.1.2 Message Switching

Entire **message is stored and forwarded** node-by-node (**store-and-forward**); no dedicated path. High delay; each node needs large storage. Largely obsolete (e.g., old telegraph).

5.1.3 Packet Switching

Message is split into **packets** routed independently or along a set-up virtual circuit.

Approach	Type	Connection	Example
Datagram	Connectionless	No setup; each packet independent, may arrive out of order	IP / Internet
Virtual Circuit (VC)	Connection-oriented	Setup path first; all packets follow same path, ordered	X.25, Frame Relay, ATM, MPLS

CIRCUIT: setup ==> dedicated path held entire call ==> teardown
 DATAGRAM: P1 ---\ /--> R packets routed
 P2 ----> ROUTERS --> R independently,
 P3 ---/ \--> R may reorder

Feature	Circuit	Datagram	Virtual Circuit
Path	Dedicated, fixed	Per-packet, varies	Fixed for session
Setup phase	Yes	No	Yes
Ordering	Guaranteed	Not guaranteed	Guaranteed
Bandwidth use	Reserved (can waste)	On demand	On demand
Addressing	Once at setup	Full address each packet	VC number each packet

MCQ Trap: In **datagram** switching each packet carries the **full destination address** and may take different paths; in **virtual circuit** each packet carries a short **VCI (virtual circuit identifier)**. The **Internet (IP)** uses the **datagram** approach.

6. Data Link Layer (DLL)

The DLL provides **node-to-node (hop-to-hop)** delivery. Two sublayers: **LLC (Logical Link Control, 802.2)** and **MAC (Medium Access Control)**.

6.1 DLL Functions and Framing

Core functions: **framing, physical (MAC) addressing, flow control, error control, access control.**

Framing packs bits into distinguishable **frames**. Methods:

- **Character/Byte count** — first field states the frame's length (fragile: one error desyncs all).
- **Byte/Character stuffing** — flag bytes (e.g., **FLAG**) mark frame boundaries; an **escape byte (ESC)** is stuffed before any FLAG/ESC appearing in data.
- **Bit stuffing** — a **flag 01111110** marks boundaries; the sender inserts a **0 after five consecutive 1s** in data; the receiver removes it.

Bit stuffing example:

```
Data      : 0 1 1 1 1 1 1 0
Stuffed   : 0 1 1 1 1 1 0 1 0   (a 0 inserted after five 1s)
Flag      : 0 1 1 1 1 1 1 0   (never altered -> boundary)
```

MCQ Trap: In **bit stuffing** a **0** is stuffed after **five consecutive 1s** (not six). The HDLC flag is **01111110**.

6.2 Error Detection and Correction

6.2.1 Types of Errors

- **Single-bit error** — only **one bit** changes.
- **Burst error** — **two or more bits** change (most common in serial transmission); burst length = first to last changed bit.

6.2.2 Parity Check

- **Simple (single) parity** — adds **1 parity bit** so total 1s are even (even parity) or odd; detects **all odd-numbered** bit errors, **fails for even** errors. Cannot correct.
- **Two-dimensional parity** — parity over rows and columns; detects most burst errors, can correct **single-bit** errors.

6.2.3 Checksum

Used at the **transport/network** layer. Sender adds segments using **1's complement**, sends the complemented sum; receiver re-adds — result of all 1s means **no error**. Detects errors but is weaker than CRC.

6.2.4 Cyclic Redundancy Check (CRC)

Powerful **detection** method using **binary (modulo-2) polynomial division**.

- Sender appends **r = degree of generator polynomial G(x)** zero bits, divides by **G(x)**, and the **remainder (CRC)** replaces those zeros.
- Receiver divides the received frame by **G(x)**; **zero remainder = no error**.
- **CRC-32** is used in **Ethernet (FCS)**.

```
CRC: data=1001, generator G=1011 (r=3) -> append 000
      1001000 / 1011 (modulo-2 / XOR division)
      remainder = 110 -> transmitted frame = 1001 110
Receiver divides 1001110 by 1011 -> remainder 000 = OK
```

MCQ Trap: CRC uses **XOR (modulo-2)** arithmetic, not normal subtraction. If the generator is **degree r**, append **r zero bits** and the CRC is **r bits** long.

6.2.5 Hamming Code (Error Correction)

Hamming code detects and **corrects single-bit errors** using **parity bits at power-of-2 positions** (1, 2, 4, 8, ...).

- Number of redundant bits **r** for **m** data bits satisfies $2^r \geq m + r + 1$.
- **Hamming distance** = number of differing bit positions; **minimum Hamming distance d_min** of a code determines power:
- Detect up to **(d_min - 1)** errors.
- Correct up to **⌊(d_min - 1)/2⌋** errors.
- For single-error correction, **d_min = 3** is required (correct **t** errors needs $d_{\min} \geq 2t + 1$; detect **d** errors needs $d_{\min} \geq d + 1$).

For m = 7 data bits: find r so $2^r \geq m + r + 1$
 r=4 $\rightarrow 2^4 = 16 \geq 7+4+1 = 12$ TRUE $\rightarrow r = 4$ parity bits
 Parity bits sit at positions 1, 2, 4, 8 (powers of 2)

MCQ Trap: Hamming code corrects a **single-bit** error. To **correct t** errors you need $d_{\min} \geq 2t+1$; to only **detect d** errors you need $d_{\min} \geq d+1$. Parity bits go at **powers of 2** positions.

6.3 Flow Control and Error Control (ARQ)

Flow control prevents a fast sender from overwhelming a slow receiver. **Error control = error detection + retransmission (ARQ = Automatic Repeat reQuest).**

6.3.1 Stop-and-Wait

Sender sends **one frame** then **waits** for an ACK before sending the next. Simple but inefficient on long links.

- **Efficiency $\eta = 1 / (1 + 2a)$** where **a = Tp / Tt** (propagation time / transmission time).
- **Throughput = $\eta \times \text{Bandwidth}$.**

6.3.2 Sliding Window Protocols

Sender may transmit up to **W** frames before needing an ACK; the **window slides** as ACKs arrive.

- **Efficiency $\eta = N / (1 + 2a)$** (capped at 1), where **N** = window size.
- **Optimal window $N = 1 + 2a$** gives 100% utilization.

6.3.3 Go-Back-N ARQ (GBN)

- **Sender window = up to $2^m - 1$** (m = sequence-number bits); **receiver window = 1**.
- On a lost/erroneous frame, sender **retransmits that frame and ALL subsequent** frames (cumulative ACK).
- Uses **cumulative acknowledgements**.

6.3.4 Selective Repeat ARQ (SR)

- **Sender window = receiver window = $2^{(m-1)}$.**
- Only the **specific lost frame** is retransmitted; receiver **buffers** out-of-order frames.
- Uses **selective (individual) ACKs**; needs sorting/buffering at receiver.

Feature	Stop-and-Wait	Go-Back-N	Selective Repeat
Sender window	1	$2^m - 1$	$2^{(m-1)}$
Receiver window	1	1	$2^{(m-1)}$
Retransmission	The frame	Frame + all after	Only lost frame
Receiver buffering	No	No	Yes
ACK type	Individual	Cumulative	Selective

Feature	Stop-and-Wait	Go-Back-N	Selective Repeat
Efficiency	Lowest	Medium	Highest

- **Piggybacking** — carrying an **ACK inside an outgoing data frame** to save bandwidth (used in bidirectional links).

MCQ Trap: For **m-bit** sequence numbers: **GBN** max sender window = $2^m - 1$; **SR** max window = $2^{(m-1)}$. Go-Back-N receiver window is always **1**.

6.4 Medium Access Control (MAC) Sublayer

The MAC sublayer decides **who transmits when** on a shared medium (the multiple-access problem).

6.4.1 Channel Allocation Categories

- **Random Access** — ALOHA, CSMA/CD, CSMA/CA (contention-based).
- **Controlled Access** — Reservation, Polling, Token Passing.
- **Channelization** — FDMA, TDMA, CDMA (divide the channel).

6.4.2 ALOHA

Type	Vulnerable Period	Max Efficiency (throughput)
Pure ALOHA	$2 \times T_{fr}$	18.4% ($1/2e$) at G = 0.5
Slotted ALOHA	$1 \times T_{fr}$	36.8% ($1/e$) at G = 1

- **Pure ALOHA** — transmit whenever ready; collision if any overlap.
- **Slotted ALOHA** — transmit only at **slot boundaries**, halving the vulnerable time and **doubling** throughput.

6.4.3 CSMA and Variants

CSMA (Carrier Sense Multiple Access) = "listen before talk".

- **Persistence methods:** **1-persistent** (send immediately when idle), **non-persistent** (wait random then resense), **p-persistent** (send with probability p).
- **CSMA/CD (Collision Detection)** — "listen while talking"; used in **wired Ethernet (IEEE 802.3)**; aborts on collision and uses **binary exponential backoff**. Needs **minimum frame size** so collisions are detected (frame time $\geq 2 \times$ propagation).
- **CSMA/CA (Collision Avoidance)** — used in **wireless (IEEE 802.11 / Wi-Fi)**; avoids collisions using **IFS, contention window, and RTS/CTS** (collisions can't be reliably detected wirelessly).

MCQ Trap: **CSMA/CD = wired Ethernet, CSMA/CA = Wi-Fi (wireless)**. Wireless uses **avoidance** because the hidden-terminal problem makes detection unreliable.

6.4.4 Controlled Access and Channelization

- **Reservation** — stations reserve slots in advance.
- **Polling** — a **primary** station invites **secondary** stations to send (poll/select).
- **Token passing** — a **token** circulates; only the token holder transmits (Token Ring/Bus).
- **FDMA** — different **frequencies**; **TDMA** — different **time slots**; **CDMA** — different **codes** (all share band/time, separated by orthogonal codes).

6.5 IEEE 802 Standards

Standard	Technology	Access Method
IEEE 802.1	Bridging / VLAN (802.1Q)	—
IEEE 802.2	LLC sublayer	—
IEEE 802.3	Ethernet	CSMA/CD
IEEE 802.4	Token Bus	Token passing on a bus
IEEE 802.5	Token Ring	Token passing on a ring
IEEE 802.11	Wireless LAN (Wi-Fi)	CSMA/CA
IEEE 802.15	WPAN — Bluetooth (802.15.1), Zigbee (802.15.4)	—
IEEE 802.16	WiMAX (broadband wireless)	—

- **Ethernet families: Standard 10 Mbps, Fast Ethernet 100 Mbps, Gigabit Ethernet 1 Gbps, 10-Gigabit 10 Gbps.**
- **Wi-Fi:** 802.11b (11 Mbps), 802.11g (54 Mbps), 802.11n, 802.11ac, 802.11ax (Wi-Fi 6).

MCQ Trap: 802.3 = CSMA/CD (Ethernet), 802.4 = Token Bus, 802.5 = Token Ring, 802.11 = Wi-Fi (CSMA/CA), 802.15 = Bluetooth/WPAN, 802.16 = WiMAX.

6.6 Wired and Wireless LAN

6.6.1 Ethernet Frame Format (IEEE 802.3)

+-----+	+-----+	+-----+	+-----+	+-----+	+-----+	+-----+	+-----+
Preamble	SFD	Dest	Source	Len/Typ	Data	FCS	
7 bytes	1 B	MAC 6B	MAC 6B	2 B	46-1500 B	4 B	
+-----+	+-----+	+-----+	+-----+	+-----+	+-----+	+-----+	+-----+

- **Preamble** = 7 bytes (sync), **SFD** = 1 byte (10101011).
- **Min frame = 64 bytes, Max frame = 1518 bytes** (excluding preamble/SFD); **MTU/payload max = 1500 bytes**, min data = **46 bytes** (padded).
- **FCS** = 4-byte **CRC-32**.

6.6.2 MAC Addressing

- **MAC (physical) address = 48 bits = 6 bytes = 12 hex digits**, e.g., **00:1A:2B:3C:4D:5E**.
- First **24 bits = OUI** (manufacturer), last 24 bits = device-specific.
- **Broadcast MAC = FF:FF:FF:FF:FF:FF**.

6.6.3 Bluetooth and Wi-Fi

- **Bluetooth (802.15.1)** — short-range WPAN, **piconet** (1 master + up to **7 active** slaves), uses **FHSS** at **2.4 GHz**.
- **Wi-Fi (802.11)** — WLAN, infrastructure (AP) or ad-hoc mode, uses **CSMA/CA**.

MCQ Trap: A Bluetooth **piconet** has **1 master and up to 7 active slaves** (255 parked). MAC address is **48-bit**, IPv4 is **32-bit** — do not confuse them.

7. Network Layer and IP Addressing

The Network layer provides **host-to-host (source-to-destination)** delivery using **logical addressing** and **routing**.

7.1 Network Layer Functions

Functions: **logical addressing (IP)**, **routing**, **packetizing/fragmentation**, and **congestion control**. IP service is **connectionless and best-effort** (no guarantee of delivery/order).

- **Routing** — determining the path (control plane).
- **Forwarding** — moving a packet to the correct output (data plane).

7.2 IPv4 Addressing

An **IPv4 address** is **32 bits**, written as **4 octets in dotted-decimal** (e.g., **192.168.1.1**), each octet **0-255**. Total space = $2^{32} \approx 4.29$ billion addresses.

7.2.1 Classful Addressing

Class	Leading bits	First octet	Default Mask	#Networks	#Hosts/net
A	0	0-127	/8 (255.0.0.0)	$2^7 - 2 = 126$	$2^{24} - 2$
B	10	128-191	/16 (255.255.0.0)	2^{14}	$2^{16} - 2$
C	110	192-223	/24 (255.255.255.0)	2^{21}	$2^8 - 2 = 254$
D	1110	224-239	—	Multicast	—
E	1111	240-255	—	Reserved/experimental	—

- Subtract **2** for usable hosts (network address + broadcast address are reserved).

7.2.2 Special Addresses

Address / Range	Meaning
127.0.0.0/8 (127.0.0.1)	Loopback (localhost)
255.255.255.255	Limited broadcast
10.0.0.0/8	Private (Class A)
172.16.0.0/12	Private (Class B)
192.168.0.0/16	Private (Class C)
169.254.0.0/16	APIPA (auto-config when DHCP fails)
0.0.0.0	"This host" / default route

MCQ Trap: 127.0.0.1 = loopback, 169.254.x.x = APIPA (DHCP failure), classes **D (224-239) = multicast** and **E (240-255) = reserved**. Class C gives only **254** usable hosts per network.

7.3 Subnetting and Supernetting

- **Subnet mask** — separates **network** and **host** bits; written in dotted-decimal or **CIDR /n** notation.
- **Subnetting** — borrowing **host bits** to create more, smaller networks.
- **Supernetting** — combining several networks into a larger block (opposite of subnetting).
- **CIDR (Classless Inter-Domain Routing)** — drops class boundaries; uses **/n prefix** (e.g., **192.168.1.0/26**).

Subnetting key formulas (n = borrowed bits, h = host bits left):

Number of subnets = 2^n
 Hosts per subnet = $2^h - 2$
 Block size (increment) = $256 - (\text{mask octet value})$
 Network address = first address (host bits all 0)
 Broadcast address = last address (host bits all 1)

Example: 192.168.1.0/26 (mask 255.255.255.192)

host bits = $32 - 26 = 6$ → hosts = $2^6 - 2 = 62$

block size = $256 - 192 = 64$

Subnets: .0 .64 .128 .192

For .0 subnet: network=.0 first host=.1 last host=.62 broadcast=.63

- **FLSM (Fixed-Length Subnet Mask)** — all subnets equal size.

- **VLSM (Variable-Length Subnet Mask)** — subnets of different sizes (efficient).

MCQ Trap: Usable hosts per subnet = $2^h - 2$ (subtract network + broadcast). A /30 has **2 usable hosts** (common for point-to-point links). The number of subnets from borrowing **n** bits = 2^n .

7.4 IPv4 Datagram and Supporting Protocols

7.4.1 IPv4 Header

The IPv4 header is **20 bytes minimum, 60 bytes maximum** (with options).

Field	Size	Purpose
Version	4 bits	4 for IPv4
IHL	4 bits	Header length (in 32-bit words)
Type of Service / DSCP	8 bits	QoS / priority
Total Length	16 bits	Header + data (max 65,535 bytes)
Identification	16 bits	Fragment grouping ID
Flags	3 bits	DF (don't fragment), MF (more fragments)
Fragment Offset	13 bits	Position of fragment (×8 bytes)
TTL	8 bits	Time To Live (hop limit; decremented per router)
Protocol	8 bits	ICMP=1, IGMP=2, TCP=6, UDP=17, OSPF=89
Header Checksum	16 bits	Error check on header only
Source / Dest IP	32 bits each	Addresses
Options	0-40 bytes	Optional

7.4.2 Fragmentation

If a datagram exceeds a link's **MTU**, it is split into **fragments** using **Identification**, **Flags (MF)**, and **Fragment Offset**; reassembly happens at the **destination**. **TTL** prevents infinite loops (router discards at TTL=0 and sends ICMP).

7.4.3 ARP, RARP, ICMP, IGMP

Protocol	Maps / Function	Layer
ARP	IP → MAC (logical to physical)	L2/L3
RARP	MAC → IP (legacy; replaced by DHCP/BOOTP)	L2/L3
ICMP	Error/diagnostic messages (ping, traceroute)	L3
IGMP	Multicast group management	L3

- **ARP** broadcasts "who has IP X?"; the owner replies with its MAC (cached in ARP table).
- **ICMP** carries errors like **Destination Unreachable, Time Exceeded, Echo Request/Reply** (ping).

7.4.4 NAT and DHCP

- **NAT (Network Address Translation)** — maps **private** IPs to **public** IPs; **PAT (overloading)** uses port numbers to share one public IP among many hosts. Conserves IPv4 addresses.
- **DHCP (Dynamic Host Configuration Protocol)** — auto-assigns IP, mask, gateway, DNS via the **DORA** process: **Discover → Offer → Request → Acknowledge**. Server port **67**, client **68** (UDP).

MCQ Trap: ARP = IP→MAC, RARP/DHCP = MAC→IP. DHCP uses UDP ports **67 (server)** and **68 (client)** and follows **DORA**. ping/traceroute use **ICMP**, not TCP/UDP.

7.5 IPv6

IPv6 address = 128 bits, written as **8 groups of 16-bit hex** separated by colons, e.g., **2001:0db8:0000:0000:0000:ff00:0042:8329**. Space = **2¹²⁸**.

- **Header is a fixed 40 bytes** (vs IPv4's variable 20-60).
- **No header checksum, no fragmentation by routers** (only the source fragments), no broadcast.
- Address types: **Unicast, Multicast, Anycast** (no broadcast in IPv6).
- Zero-compression: consecutive zero groups → **::** (used once).

Feature	IPv4	IPv6
Size	32-bit	128-bit
Notation	Dotted decimal	Colon hex
Header	20-60 B, variable	40 B fixed
Checksum	Yes	No
Broadcast	Yes	No (uses multicast/anycast)
Fragmentation	Routers + source	Source only
Address config	Manual / DHCP	SLAAC / DHCPv6

- **Transition mechanisms: Dual stack** (run both), **Tunneling** (encapsulate IPv6 in IPv4), **Translation (NAT64)**.

MCQ Trap: IPv6 has **NO broadcast** and **NO header checksum**, and routers **do not fragment** IPv6. The IPv6 base header is a **fixed 40 bytes**.

7.6 Routing

7.6.1 Routing vs Forwarding; Static vs Dynamic

- **Static routing** — manually configured; good for small/stable networks.
- **Dynamic routing** — routers exchange info via routing protocols and adapt automatically.

7.6.2 Routing Algorithm Families

Algorithm	Basis	Protocol(s)	Metric
Distance Vector	Bellman-Ford	RIP, IGRP	Hop count
Link State	Dijkstra	OSPF, IS-IS	Cost (bandwidth)
Path Vector	Path of AS	BGP	Policy/AS-path

- **Distance Vector** — each router shares its **whole table with neighbours** periodically ("routing by rumour"); suffers the **count-to-infinity** problem. **RIP**: metric = **hop count, max 15 hops (16 = infinity/unreachable)**, updates every **30 s**, uses **UDP port 520**.
- Fixes: **split horizon, poison reverse, hold-down timers**.
- **Link State** — each router floods **link-state advertisements (LSAs)** to **all** routers, builds a full topology map, then runs **Dijkstra's shortest path**. **OSPF** is hierarchical (areas), uses **cost**, runs directly over IP (**protocol 89**), converges fast.
- **Path Vector** — **BGP** is the inter-domain (between AS) protocol of the Internet; uses **TCP port 179** and AS-path/policies.
- **Flooding** — send incoming packet on **every** outgoing line except the source; reliable but wasteful (controlled by hop count/sequence numbers).

Feature	Distance Vector	Link State
Algorithm	Bellman-Ford	Dijkstra
Knowledge	Only neighbours	Entire topology

Feature	Distance Vector	Link State
Updates	Full table, periodic	LSAs on change
Convergence	Slow	Fast
Count-to-infinity	Yes	No
Example	RIP	OSPF

MCQ Trap: RIP = Distance Vector, hop count, max 15 hops, 16 = infinity; OSPF = Link State, Dijkstra, cost metric; BGP = Path Vector, TCP 179. Distance vector = Bellman-Ford and suffers count-to-infinity.

8. Transport Layer

The Transport layer provides **process-to-process (end-to-end)** delivery, identifying applications via **port numbers**.

8.1 Transport Layer Services

- **Process-to-process delivery** using **port numbers** (16-bit, **0-65535**).
- **Socket = IP address + Port number** (uniquely identifies an endpoint).
- **Multiplexing/Demultiplexing** — many processes share the network via ports.

Port Range	Name	Examples
0-1023	Well-known	HTTP 80, HTTPS 443, DNS 53
1024-49151	Registered	Vendor apps
49152-65535	Dynamic / Ephemeral	Client temporary ports

8.2 UDP (User Datagram Protocol)

UDP is **connectionless, unreliable, fast**; no handshake, ordering, or flow control. Header is a fixed **8 bytes**.

UDP header (8 bytes):

```

+-----+-----+
| Source Port | Dest Port | 16 + 16 bits
+-----+-----+
| Length      | Checksum  | 16 + 16 bits
+-----+-----+
```

- Fields: **Source Port, Destination Port, Length, Checksum** (each 16 bits).
- Use cases: **DNS, DHCP, TFTP, SNMP, VoIP, streaming, online gaming** (low latency, loss tolerable).

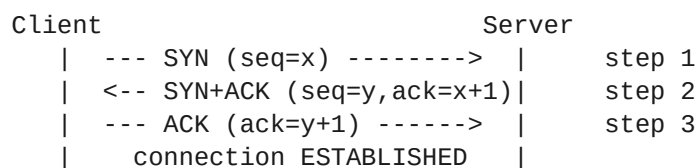
8.3 TCP (Transmission Control Protocol)

TCP is **connection-oriented, reliable, byte-stream**; provides ordered delivery, flow control, error control, and congestion control. Header **20-60 bytes**.

8.3.1 TCP Header Fields

Field	Size	Purpose
Source / Dest Port	16 bits each	Endpoints
Sequence Number	32 bits	Byte stream position
Acknowledgement Number	32 bits	Next expected byte
HLEN / Data Offset	4 bits	Header length
Flags	6 (control)	URG, ACK, PSH, RST, SYN, FIN
Window Size	16 bits	Flow control (receiver window)
Checksum	16 bits	Error check
Urgent Pointer	16 bits	With URG flag

8.3.2 Three-Way Handshake (Connection Setup)



- **Step 1:** Client sends **SYN**. **Step 2:** Server replies **SYN + ACK**. **Step 3:** Client sends **ACK**. Connection is now established.

8.3.3 Connection Termination (4-Way)

FIN → ACK → FIN → ACK — each side closes its direction independently (full-duplex close). Involves a **TIME-WAIT** state.

MCQ Trap: Connection setup is **3-way (SYN, SYN-ACK, ACK)** but termination is **4-way (FIN, ACK, FIN, ACK)**. The **SYN** flag synchronizes sequence numbers; **RST** abruptly resets.

8.4 Flow Control and Congestion Control

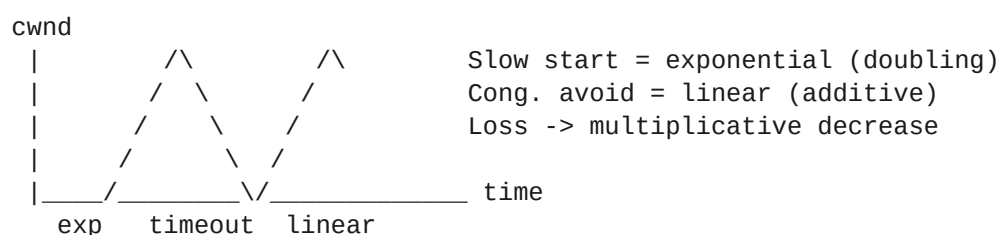
8.4.1 Flow Control

TCP uses a **sliding window** advertised by the **receiver window (rwnd)** to throttle the sender (end-to-end).

8.4.2 Congestion Control (AIMD)

TCP keeps a **congestion window (cwnd)**; effective window = **min(rwnd, cwnd)**.

- **Slow Start** — cwnd starts at 1 MSS and **doubles each RTT (exponential)** until it reaches **ssthresh**.
- **Congestion Avoidance** — beyond ssthresh, cwnd grows **linearly (additive increase, +1 MSS/RTT)**.
- **Fast Retransmit** — **3 duplicate ACKs** trigger immediate retransmission without waiting for timeout.
- **Fast Recovery** — after fast retransmit, halve cwnd and continue (avoids slow start).
- Overall policy = **AIMD (Additive Increase, Multiplicative Decrease)**.



8.4.3 Traffic Shaping — Leaky vs Token Bucket

Feature	Leaky Bucket	Token Bucket
Output rate	Constant (fixed)	Variable; allows bursts
Bursts	Smoothed/removed	Permitted (saved tokens)
Idle credit	None	Accumulates tokens
Analogy	Leaking water	Tokens added per unit time

8.4.4 QoS Basics

Quality of Service parameters: **bandwidth, delay (latency), jitter, packet loss**. Techniques: scheduling, traffic shaping, admission control, IntServ/DiffServ.

8.4.5 TCP vs UDP

Feature	TCP	UDP
Connection	Connection-oriented	Connectionless
Reliability	Reliable (ACK, retransmit)	Unreliable (best-effort)
Ordering	Ordered	No ordering
Header size	20-60 bytes	8 bytes
Speed	Slower	Faster
Flow/Congestion control	Yes	No
Use	HTTP, FTP, SMTP, SSH	DNS, DHCP, VoIP, streaming

MCQ Trap: UDP header = 8 bytes (fixed), TCP header = 20-60 bytes. TCP flags = **URG, ACK, PSH, RST, SYN, FIN**. UDP has **no congestion control**.

9. Application Layer

9.1 DNS (Domain Name System)

DNS translates **domain names** → **IP addresses**; it is a **distributed, hierarchical** database using **UDP/TCP port 53** (UDP for queries, **TCP for zone transfers** and large responses).

- Hierarchy: **Root (.)** → **Top-Level Domain (TLD: .com, .org)** → **Second-level** → **host**.
- **Recursive resolution** — the resolver does all the work and returns the final answer.
- **Iterative resolution** — each server returns a **referral** to the next server.

DNS Record	Meaning
A	Name → IPv4 address
AAAA	Name → IPv6 address
MX	Mail exchange server
CNAME	Alias to another name
NS	Authoritative name server
PTR	IP → name (reverse lookup)
SOA	Start of Authority (zone info)

MCQ Trap: DNS uses **port 53** — **UDP** for normal queries, **TCP** for **zone transfers**. **A** record = IPv4, **AAAA** = IPv6, **MX** = mail, **PTR** = reverse lookup.

9.2 Application Protocols and Ports

Protocol	Port	Transport	Purpose
FTP (data)	20	TCP	File data transfer
FTP (control)	21	TCP	File transfer commands
SSH	22	TCP	Secure remote login
Telnet	23	TCP	Remote login (insecure)
SMTP	25	TCP	Send/relay email (push)
DNS	53	UDP/TCP	Name resolution
DHCP	67/68	UDP	IP auto-configuration
TFTP	69	UDP	Trivial file transfer
HTTP	80	TCP	Web
POP3	110	TCP	Retrieve email (download+delete)
NTP	123	UDP	Time sync
IMAP	143	TCP	Retrieve email (sync, server-stored)
SNMP	161/162	UDP	Network management (162 = trap)
BGP	179	TCP	Inter-domain routing
HTTPS	443	TCP	Secure web (TLS/SSL)

- **HTTP** is **stateless**; methods include **GET, POST, PUT, DELETE, HEAD, OPTIONS**. State is tracked using **cookies**.
- **HTTPS = HTTP over TLS/SSL** (encryption + authentication), port **443**.
- **Email push vs pull: SMTP pushes** mail to the server; **POP3/IMAP pull** mail to the client. **IMAP** keeps mail on the **server (multi-device sync)**; **POP3** typically **downloads and deletes**.
- **Telnet (23) is insecure** (plaintext); **SSH (22)** is the encrypted replacement.

MCQ Trap: SMTP=25, POP3=110, IMAP=143; SMTP is for **sending**, POP3/IMAP for **receiving**. HTTP=80, HTTPS=443. DNS=53, DHCP=67/68, SNMP=161/162. FTP uses **20 (data)** and **21 (control)**.

Memory Trick: Sorted ports: **FTP 20/21 → SSH 22 → Telnet 23 → SMTP 25 → DNS 53 → DHCP 67/68 → TFTP 69 → HTTP 80 → POP3 110 → IMAP 143 → HTTPS 443**.

9.3 World Wide Web and Internet

- **WWW** — a system of interlinked **hypertext documents** accessed via the Internet; uses **HTTP/HTTPS** and **URLs**.
- **URL (Uniform Resource Locator)** — `protocol://host:port/path?query`, e.g., `https://www.example.com:443/index.html`.
- **Internet architecture** — a global **internetwork** of autonomous systems connected by routers; end hosts use the **TCP/IP** stack; ISPs interconnect at **IXPs**.
- **Web browser** — client that requests and renders web pages.
- **Proxy server** — intermediary that **caches** content, filters, and improves privacy/performance.
- **Cookies** — small data stored by the browser to maintain **state** (sessions, preferences) over stateless HTTP.

MCQ Trap: HTTP is stateless; cookies add state. A **proxy server caches** and filters requests. The **WWW is a service that runs over the Internet** — they are not the same thing.

10. Networking Devices and NIC

10.1 Devices and OSI-Layer Mapping

Device	OSI Layer	Function
Repeater	L1 — Physical	Regenerates/amplifies signal (extends distance)
Hub	L1 — Physical	Multiport repeater; broadcasts to all ports; one collision domain
Bridge	L2 — Data Link	Connects LAN segments; filters by MAC ; breaks collision domains
Switch	L2 (L3 switch = L3)	Multiport bridge; MAC-based forwarding ; each port = own collision domain; VLAN
Router	L3 — Network	Forwards packets by IP ; connects different networks; breaks broadcast domains
Gateway	L7 / all layers	Protocol converter between dissimilar networks
Modem	L1 — Physical	Modulate/demodulate (digital ↔ analog)
Brouter	L2 + L3	Bridge + router hybrid
NIC	L1-L2	Network Interface Card ; holds the burned-in MAC address

APPLICATION ----- Gateway
 ...
 NETWORK (L3) ----- Router, L3 Switch, Brouter
 DATA LINK (L2) --- Switch, Bridge, NIC
 PHYSICAL (L1) ---- Hub, Repeater, Modem, Cables

- **Hub vs Switch:** a **hub** shares bandwidth and creates **one collision domain** (half-duplex); a **switch** gives each port dedicated bandwidth, **separate collision domains**, and full-duplex.
- **Switch vs Router:** a **switch** forwards by **MAC** within a LAN (one broadcast domain unless VLANs); a **router** forwards by **IP** and **separates broadcast domains**.
- **VLAN** — a switch feature creating **logical LANs** that segment broadcast domains (802.1Q tagging).
- **NIC** — provides the physical connection and the unique **48-bit MAC address**.

MCQ Trap: Hub/Repeater = L1, Switch/Bridge/NIC = L2, Router = L3, Gateway = L7/all. A **switch separates collision domains** but **not broadcast domains**; a **router separates both**. Number of collision domains = number of switch ports; a hub = **1** collision domain.

Concept	Hub	Switch	Router
Collision domains	1	per port	per port
Broadcast domains	1	1 (per VLAN)	per port
Forwarding by	None (broadcast)	MAC	IP
Layer	1	2	3

11. Network Security (Overview)

11.1 Security Goals — CIA Triad

- **Confidentiality** — only authorized parties can read data (encryption).
- **Integrity** — data is **not altered** in transit (hashes/MAC).
- **Availability** — services remain **accessible** (anti-DoS, redundancy).

11.2 Cryptography

Type	Keys	Examples	Speed
Symmetric	Same key for encrypt/decrypt	DES (56-bit), 3DES, AES (128/192/256)	Fast
Asymmetric (Public-key)	Key pair (public + private)	RSA, Diffie-Hellman, ECC	Slow
Hashing	None (one-way)	MD5 (128-bit), SHA-1 (160-bit), SHA-256	Fast

- **Symmetric** problem: secure **key distribution**; **asymmetric** solves it but is slower → hybrid (e.g., TLS uses asymmetric to exchange a symmetric session key).
- **Diffie-Hellman** = key **exchange** (not encryption).

11.3 Digital Signatures

A **digital signature** provides **authentication, integrity, and non-repudiation**. The sender **signs a message hash with their PRIVATE key**; the receiver **verifies with the sender's PUBLIC key**.

MCQ Trap: For **confidentiality** encrypt with the **receiver's public** key; for a **digital signature** sign with the **sender's private** key. **Hashing ≠ encryption** (hashing is one-way, irreversible).

11.4 Firewalls, VPN, and Attacks

- **Firewall** — filters traffic by rules; types: **packet-filter, stateful inspection, application/proxy gateway**.
- **VPN** — secure **encrypted tunnel** over a public network (e.g., **IPSec**, SSL/TLS VPN).
- **Common attacks:** **DoS/DDoS** (availability), **Man-in-the-Middle, phishing, spoofing (IP/MAC), sniffing/eavesdropping, replay, SQL injection, malware**.

MCQ Trap: A **firewall** controls traffic flow but does not encrypt by itself; a **VPN** provides the encrypted tunnel. **DoS** attacks target **availability**, not confidentiality.

12. Quick Revision and Exam Pointers

12.1 Must-Memorize Numbers

- OSI = **7 layers**; TCP/IP = **4 layers**.
- IPv4 = **32-bit**; IPv6 = **128-bit**; MAC = **48-bit**; Port = **16-bit**.
- IPv4 header = **20-60 B**; IPv6 header = **40 B fixed**; TCP = **20-60 B**; UDP = **8 B**.
- Ethernet: min frame **64 B**, max **1518 B**, MTU **1500 B**; FCS = **CRC-32 (4 B)**.
- Pure ALOHA = **18.4%**, Slotted ALOHA = **36.8%**.
- RIP: hop count, **max 15**, infinity **16**, updates every **30 s**, **UDP 520**.
- Stop-and-Wait efficiency = **$1/(1+2a)$** ; Sliding window = **$N/(1+2a)$** .
- GBN window = **$2^m - 1$** ; SR window = **2^{m-1}** .

12.2 Key Difference Tables to Revise

- **OSI vs TCP/IP** — 7 vs 4 layers; ISO vs DoD.
- **TCP vs UDP** — reliable/CO vs unreliable/CL; 20-60 B vs 8 B header.
- **Hub vs Switch vs Router** — L1 vs L2 vs L3; collision/broadcast domains.
- **Circuit vs Packet switching** — dedicated path vs independent packets.
- **IPv4 vs IPv6** — 32-bit vs 128-bit; broadcast vs no broadcast.
- **Distance Vector vs Link State** — Bellman-Ford/RIP vs Dijkstra/OSPF.
- **Go-Back-N vs Selective Repeat** — retransmit all vs only lost.
- **Leaky vs Token Bucket** — fixed rate vs allows bursts.

12.3 Numerical Practice Checklist

- Subnetting: number of subnets 2^n , hosts $2^h - 2$, block size $256 - \text{mask octet}$.
- Sliding-window efficiency and bandwidth-delay product $\text{BDP} = \text{bandwidth} \times \text{RTT}$.
- CRC: append r zero bits, modulo-2 divide by degree- r generator.
- Hamming: redundant bits $2^r \geq m + r + 1$; correct t errors needs $d_{\min} \geq 2t + 1$.
- Channel capacity: **Nyquist** $C = 2B \cdot \log_2(L)$; **Shannon** $C = B \cdot \log_2(1 + S/N)$.
- PCM: sampling rate $\geq 2 \cdot f_{\max}$; voice = **64 kbps**.

Memory Trick: Layer mnemonics — bottom-up "Please Do Not Throw Sausage Pizza Away"; PDUs "Some People Fear Birthdays" (Segment, Packet, Frame, Bit). Devices: **H**ub/**R**epeater = L1, **S**witch/**B**ridge = L2, **R**outer = L3, **G**ateway = L7.

Shortcut: If a question gives S/N use **Shannon**; if it gives levels L use **Nyquist**. If it mentions **hop count**, think **RIP/Distance Vector**; **cost/Dijkstra**, think **OSPF/Link State**. **CSMA/CD** = wired, **CSMA/CA** = wireless.